Alejandro Diaz Fontalva

SOFTWARE ENGINEER (C#)
C# / .NET DEVELOPER

ABOUT ME

Software Engineer specialized in C# and .NET development, building desktop applications with WPF and Windows Forms, focused on industrial and data-driven software systems.

CONTACT

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SKILLS & TOOLS

Languages
C#, JavaScript, TypeScript, SQL, Python, C/C++, PHP

Frontend
Angular, HTML, CSS

Frameworks & Libraries
.NET, WPF, Windows Forms

Architecture & Patterns
OOP, SOLID, MVVM

Tools
Git, Visual Studio, Visual Studio Code, SQL Server

Domains
Software Development · Desktop Applications · Industrial Systems

LANGUAGES

French - Fluent (Professional)
English - Fluent (Professional)
Spanish - Native

HOBBIES

Game dev. Tennis
Travelling Skiing

EDUCATION

Master of Mechatronics Engineering 2014
E.T.S.I. Universidad de Málaga | Málaga

Bachelor of Electronics Engineering 2012
E.T.S.I. Universidad de Málaga | Málaga

TRAINING & CERTIFICATIONS

Object-Oriented Design Patterns in C# 2025
Online Course

Modern JavaScript & Frontend Development (Angular, Node.js, Astro) In progress
Udemy | Online

ASP.NET Core & Blazor Training In progress
Udemy | Online

Complete C# Masterclass 2024
Udemy | Online

TwinCAT Motion & TwinSAFE Trainings 2023-2024
Beckhoff Automation | Yverdon les Bains

TwinCAT 3 HMI & Basis Training 2018
Beckhoff Automation | Yverdon les Bains

LabVIEW Embedded Control and Monitoring 2018
National Instruments | CERN

RELEVANT ENGINEERING EXPERIENCE

Automation Engineer – *L.E.S.S. SA, Renens* (Aug 2022 – Aug 2024)

- Developed and maintained automation software for production lines using Beckhoff PLCs; designed and implemented HMIs.
- Managed software team (2 engineers) and coordinated version control with Git.
- Contributed to all phases of a new production line design and implementation, resulting in >100 products from 20+ different work orders and automatic production of a new product type.

Industrial IT Engineer – *ETEL S.A., Môtiers* (Nov 2020 – Jul 2022)

- Developed .NET desktop applications (C#, Windows Forms, WPF) for testing Motion Ccontrol systems on production benches.
- Provided technical support, implemented new tests/features, resolved bugs, and managed Git version control.
- Supported >10 tests/day, enabling delivery of 100+ products to clients.

Controls & Automation Eng. – *CERN, Geneva* (Feb 2018 – Jan 2020)

- Developed front-end and back-end PLC control software (Beckhoff/TwinCAT 2 & 3) for accelerator systems; implemented HMIs.
- Coordinated with electrical technicians and tracked HW orders.
- Co-author of a scientific paper presenting the PLC control system for SPS RF upgrade.

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EARLY ENGINEERING EXPERIENCE

Real-Time Software Engineer – *CERN, Geneva* (Jul 2016 – Jul 2017)

- Developed LabVIEW software for data acquisition and HTS magnet testing; integrated FPGA and RT control software in cRIO systems.
- Performed post-mortem data analysis in Matlab.
- Co-author of scientific paper on HTS dipole magnet testing.

Electronic Engineer – *CERN, Geneva* (Jul 2015 – Jul 2016)

- Developed Arduino-based software for sensor calibration in robotic applications and PHP scripts for database management (Oracle/MySQL).
- Migrated robotic arm communication from native protocol to CANOpen.

Electrical Engineer – *Mijas Town Hall, Málaga* (Dec 2014 – Apr 2015)

- Developed indoor/outdoor lighting and low-voltage electrical installation projects for public buildings and roads.

SOFTWARE & WEB DEV. PROJECTS

Development of software applications across desktop and web environments using C#, .NET and modern frontend technologies, with focus on clean architecture, modular design and user interaction systems.

Technical contributions include:

- Object-oriented architectures (state machines, modular components, event-driven systems).
- WPF MVVM desktop CRUD application with async/await and SQL Server integration.
- Web development: WordPress + WooCommerce site and Angular SPA (in progress).
- UI logic, interaction systems, animations and input handling.
- Real-time interaction systems (movement controllers, ray-based interactions, collision logic).
- Data persistence, configuration handling, debugging and performance optimization.
- Version control with Git and development workflows.

Technologies:
C#, .NET, WPF (MVVM), SQL Server, JavaScript, TypeScript, Angular (in progress), HTML/CSS, Git, Visual Studio, WordPress, WooCommerce, design patterns.

PUBLICATIONS

- J. Oliveira et al., "**The PLC control system for the RF upgrade of the Super Proton Synchrotron**", ICALEPCS19, New York, Oct. 2019. (*co-author*)
- J. van Nugteren et al., "**Powering of an HTS dipole insert-magnet operated standalone in helium gas between 5 and 85 K**", IEEE Trans. Appl. Supercond., vol. 27, no. 4, Jun. 2017. (*co-author*)
- "**Highly efficient PWM strategy over FPGA**", Electronics Letters, 20th Nov. 2008, Vol. 44, No. 24. (*co-author*)